

Improving Dynamic Task Performance with Distributional Reinforcement Learning: A Real-Time Fruit Slicing Control Case Study

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Abstract - Traditional scalar-based reinforcement learning (RL) methods like Advantage Actor-Critic (A2C) estimate only the expected return, which overlooks the full return distribution. This limitation often leads to suboptimal and unstable policies, especially in high-speed, continuous action spaces. To address these limitations, this paper proposes a Distributional A2C (DA2C) algorithm inspired by the principles of Distributional Soft Actor-Critic (DSAC). Instead of relying on scalar value estimation, proposed DA2C models return distributions as Gaussian, allowing the algorithm to learn the mean and variance of returns. This representation allows the agent to make more accurate value estimates, reducing the risk of over/under estimation. Consequently, policy updates become more stable, precise, and resilient to environmental variability. The proposed method is evaluated using a real-time simulated fruit-cutting task, which is a high-speed control problem that requires quick and precise decision-making. Simulation results show that the proposed DA2C outperforms the traditional A2C baseline across key metrics. Notably, it achieves higher slicing precision, faster response, and enhanced policy robustness, showcasing the benefits of incorporating distributional value estimation in complex, time-sensitive RL applications.

Keywords - Advantage Actor-Critic (A2C); Distributional Advantage Actor-Critic (DA2C); Distributional Reinforcement Learning; Distributional Soft Actor-Critic (DSAC); Fruit Slicing Application; Policy Optimization; Reinforcement Learning (RL).

I. INTRODUCTION

Actor-Critic methods represent one of the most widely adopted paradigms in reinforcement learning (RL) [1], [2], balancing the stability of value-based approaches with the flexibility of policy-based methods. Standard algorithms such as Advantage Actor-Critic (A2C) [3]-[7], Proximal Policy Optimization (PPO) [8], and Soft Actor-Critic (SAC) [9]-[12] have been applied in robotics, games, and decision-making. However, these algorithms typically employ a scalar value function that estimates only the expected return.

While sufficient in many low-speed or discrete tasks, scalar estimators can be inadequate for high-speed continuous control, where uncertainty and variability in returns significantly affect action quality. For instance, in real-time fruit slicing tasks, the agent must react within milliseconds, accounting not only for the mean outcome of an action but

also for the distribution of possible outcomes (variance and tail probabilities). Ignoring such distributional information can lead to miscalibrated actions: the agent may over-commit to actions that appear favorable in expectation but are unreliable under temporal or environmental noise.

Distributional reinforcement learning (DistRL) addresses this limitation by modeling the full return distribution rather than its expectation. Methods such as C51, Quantile Regression DQN (QR-DQN), and Distributional Soft Actor-Critic (DSAC) [13], [14] have demonstrated that capturing return variance improves stability, risk sensitivity, and policy robustness. However, most prior work has been limited to off-policy settings (e.g., DSAC) or discrete control benchmarks. The integration of distributional critics into simple, on-policy actor-critic pipelines, such as A2C, remains underexplored, despite their importance in settings where sample efficiency, stability, and real-time constraints matter.

In this work, we propose the Distributional Advantage Actor-Critic (DA2C), an on-policy actor-critic architecture that incorporates distributional value estimation into the critic network. Unlike scalar A2C, which estimates only the expected value, DA2C models the return distribution, enabling the agent to account for uncertainty in its policy updates. To further exploit this, we design Uncertainty-Aware Policy Gradient Updates, where the contribution of each advantage estimate is weighted by its predicted variance, thereby prioritizing low-uncertainty actions and reducing instability from noisy estimates. Real-time decision-making in changing environments is a major challenge for reinforcement learning (RL) algorithms, especially in domains where quick response and fine-grained control are critical. A real-time fruit slicing is one such example, where an intelligent system needs to interact with rapidly moving objects in continuous action spaces very quickly and precisely. Classical RL models, including Advantage Actor-Critic (A2C), are useful for resolving sophisticated decision-making problems. But their use of scalar value function estimation tends to lead to less than optimal performance because of the restricted representation of uncertainty and variability of value.

In standard A2C frameworks, the value function is modelled as a simple expected return and does not model the

entire distribution of potential returns. This could result in the problem of overestimation or underestimation of value and consequently affect the efficiency and stability of policy learning. In contrast, distributional RL (DRL) algorithms seek to capture the entire distribution of returns instead of a point estimate of expected value, which allows for more accurate and informative value estimates [15], [16]. Recent developments in DRL, most notably Distributional Soft Actor-Critic (DSAC), have been shown to outperform in continuous control tasks by utilizing stochastic value distributions. The contributions of this paper are threefold:

- This presents DA2C, the first distributional variant of A2C for high-speed continuous control tasks.
- Presents an uncertainty-aware policy gradient formulation that leverages critic variance to stabilize updates.
- Validates DA2C on a challenging fruit slicing environment, demonstrating superior performance over A2C in terms of slicing accuracy, response time, and robustness.

By integrating distributional learning principles into A2C, this work presents a new development to actor-critic methods, thus enabling them for real-time, continuous-action problems. This lays the groundwork for further investigation into the combination of DRL and real-world control issues.

The organization of the paper is as follows. Section II gives the proposed methodology, including the mathematical formulation of distributional value estimation in A2C. Section III discusses the results, comparing the proposed method with baseline RL models. Section IV provides a conclusion on the results, and Section V discusses the limitations and further research directions.

II. METHODOLOGY

The proposed methodology is centered on an enhanced Advantage Actor-Critic (A2C) algorithm, incorporating DRL with elements from DSAC to improve learning stability and policy robustness. The key novelty lies in distributional value function approximation, where we approximate the full return distribution instead of only the expected value, leading to better uncertainty estimation and exploration strategies. This section discusses the proposed model framework, algorithm improvements, and mathematical formulations that define our approach, and the same can be observed in Fig. 1.

A. Reinforcement Learning (RL) Framework

RL is based on the Markov Decision Process (MDP), which models decision-making in a stochastic environment. Formally, an MDP is defined as (1).

$$M = (S, A, P, R, \gamma) \quad (1)$$

Where S is the set of all possible states representing the environment, A is the set of available actions an agent can take, $P(s'|s, a)$ is the state transition probability function determining the probability of transitioning to state s' given state s and action a . $R(s, a)$ is the reward function defining the immediate reward received after taking action a in state s . $\gamma \in [0, 1]$, is the discount factor controlling the importance of future rewards. The goal in RL is to learn an optimal policy $\pi^*(a|s)$ that maximizes the expected cumulative reward as (2). Instead of directly learning the policy, we approximate the state-value function as given in (3) and action-value function in (4). Instead of learning a single scalar $V(s)$, we extend this to distributional RL, where we learn a full return distribution.

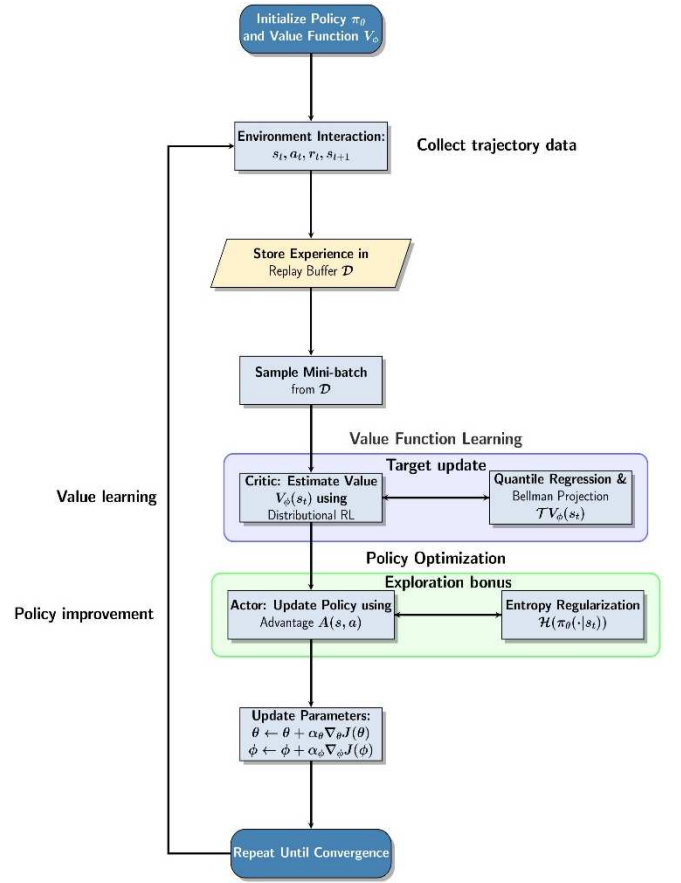


Fig. 1. Proposed model implementation flow.

$$J(\pi) = E_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right] \quad (2)$$

$$V^{\pi}(S) = E_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | s_0 = s \right] \quad (3)$$

$$Q^{\pi}(S, a) = E_{\pi} [R(s, a) + \gamma v^{\pi}(s')] \quad (4)$$

B. Advantage Actor-Critic (A2C) Algorithm

The Advantage Actor-Critic (A2C) algorithm is a policy gradient approach that unifies value-based and policy-based RL methods. The actor updates the policy $\pi_{\theta}(a|s)$, whereas the critic estimates the value function $V(s)$ and calculates the advantage function $A(s, a)$ to inform the policy updates. The ultimate goal is to maximize the expected cumulative reward given in (5).

$$J(\theta) = E_{T \sim \pi_{\theta}} \left[\sum_{t=0}^T \gamma^t R_t \right] \quad (5)$$

where T represents a trajectory sampled from the policy, γ is the discount factor, and R_t is the reward at time step t . The policy gradient update is performed using (6), where the advantage function is computed as (7) with the state-value function estimated by the critic as (8).

$$\nabla_{\theta} J(\theta) = E_{s, a \sim \pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a|s) A(s, a)] \quad (6)$$

$$A(s, a) = Q(s, a) - V(s) \quad (7)$$

$$V(s) = E[R_t + \gamma V(s')] \quad (8)$$

A2C stabilizes training with the help of multiple synchronous workers that update shared policy parameters based on their individual experience, avoiding instability caused by correlated updates. Standard A2C estimates value functions as single-point expectations, which restricts its capacity to deal with uncertainty in dynamic conditions [15].

C. Distributional Value Function Approximation in A2C

To mitigate the restrictions of one-point value estimation, we incorporate DRL into A2C by representing the state-value function $V(s)$ as a probability distribution instead of a scalar. This allows a robust management of reward uncertainty with enhanced policy stability. The proposed DA2C is not a DSAC variant but rather a distributional version of A2C. DA2C retains the on-policy training and advantage-based structure of A2C, while extending the critic to output a parametric return distribution. Key settings for the present work are as follows.

1. Actor Network: Input: state $s \in \mathbb{R}^{12}$

Architecture: MLP with two hidden layers of 256 units, ReLU activations, and the output being a Gaussian policy given in (9).

$$\pi_\theta(a|s) = \mathcal{N}(\mu_\theta(s), \sigma_\theta^2(s)) \quad (9)$$

where $\mu_\theta(s)$ is the predicted action mean and $\sigma_\theta(s)$ is the predicted standard deviation.

2. Critic Network: Input: state $s \in \mathbb{R}^{12}$

Architecture: MLP with two hidden layers of 256 units, ReLU activations, and the output being the mean and variance of the return distribution given in (10).

$$Z(s) \sim \mathcal{N}(\mu_\phi(s), \sigma^2(s)) \quad (10)$$

3. Critic Loss: From the distributional Bellman target given in (11), we obtain (12) and (13). The critic loss is given in (14).

$$Z_{tgt} = r + \gamma Z(s') \quad (11)$$

$$\mu_{tgt} = r + \gamma \mu_\phi(s') \quad (12)$$

$$\sigma_{tgt} = \gamma \sigma_\phi(s') \quad (13)$$

$$L_{critic} = (\mu_{tgt} - \mu_\phi(s))^2 + \lambda (\sigma_{tgt} - \sigma_\phi(s))^2 \quad (14)$$

4. Uncertainty-Aware Advantage: We define uncertainty-aware advantage as shown in (15).

$$\tilde{A}(s, a) = \frac{A(s, a)}{1 + \beta \sigma_\theta(s)} \quad (15)$$

Where $A(s, a) = Q(s, a) - V(s)$ is the conventional advantage, β controls uncertainty scaling (downweights uncertain critics).

5. Actor Loss: The actor loss is given by (16).

$$L_{actor} = -\mathbb{E}_{s, a \sim \pi_\theta} \tilde{A}(s, a) \log \pi_\theta(a|s) \quad (16)$$

The distributional advantage actor critic algorithm used to update the actor and critic parameters is given in Table I.

Our primary contribution lies in integrating DRL into A2C, enabling uncertainty-aware policy optimization, reward function, and reward clipping.

A reward function is generally a function mapping from a state-action pair (or state-action-next state triplet) to a scalar value (the reward). The reward scheme for the fruit-cutting problem is formulated such that it would trade off slicing accuracy, response time, and cut success. On each time step, the agent gets a reward of three major constituents: slice accuracy (SA) (calculated as the distance between the current cut position and the target position of the cut), response time (RT) (the duration to execute the cut faster response getting greater reward), and the success of the cut (rewarding successful cuts while punishing unsuccessful cuts). The reward function is given as (17). The slicing accuracy is given in (18).

$$R = (w_1 \cdot \text{slicing accuracy}) - (w_2 \cdot \text{response time}) - (w_3 \cdot \text{cut failure}) \quad (17)$$

TABLE I. DISTRIBUTIONAL ADVANTAGE ACTOR-CRITIC (DA2C)

Algorithm 1:

Input: Actor parameters θ , critic parameters ϕ
Repeat Until Convergence:
 Initialize environment, get state s
For $t \leftarrow 1$ to T , **do**
 Sample action $a \sim \pi_\theta(a|s)$;
 Execute action, observe reward r , next state s' ;
 // Critic update
 $\mu_{tgt} \leftarrow r + \gamma \mu_\phi(s')$;
 $\sigma_{tgt} \leftarrow \gamma \sigma_\phi(s')$
 $L_{critic} \leftarrow (\mu_{tgt} - \mu_\phi(s))^2 + \lambda (\sigma_{tgt} - \sigma_\phi(s))^2$;
 Update $\phi \leftarrow \phi - \eta \nabla_\phi L_{critic}$
 // Advantage estimation
 $A(s, a) \leftarrow r + \gamma \mu_\phi(s') - \mu_\phi(s)$
 $\tilde{A}(s, a) \leftarrow \frac{A(s, a)}{1 + \beta \sigma_\theta(s)}$
 // Actor update
 $L_{actor} \leftarrow -\tilde{A}(s, a) \log \pi_\theta(a, s)$
 Update $\theta \leftarrow \theta - \eta \nabla_\theta L_{actor}$;
 $s \leftarrow s'$
end
end convergence

$$SA = 1 - \frac{|\widehat{p}_{cut} - p_{gt}|}{d_{max}} \quad (18)$$

Where $\widehat{p}_{cut}$ is the predicted cut location, and p_{gt} is the ground truth ideal cut. The response time (RT) is given in (19).

$$RT = \frac{t_{exec} - t_{ideal}}{T} \quad (19)$$

It is the difference between the execution time and the ideal time taken to cut the fruit, normalized to the max cutting time. The cut failure is a parameter equal to 1 if the fruit cut fails. The weights are fixed by empirical and domain knowledge and are fixed as 1, 0.5, and 2 for w_1 , w_2 , and w_3 , respectively.

In the fruit slice application, reward clipping (20) avoids large values from destabilizing the learning process and also ensures that the rewards are limited within the range $[-\alpha, \alpha]$.

$$R_{clipped} = \text{clip}(R, -\alpha, \alpha) \quad (20)$$

Where R is the raw reward calculated from the reward function. α is a clipping threshold (1 or 10), ensuring the rewards are limited within the range $[-\alpha, \alpha]$. This ensures that extreme rewards do not disproportionately affect the agent's learning process and helps in stabilizing training, especially when combined with distributional RL models like DA2C.

III. SIMULATION RESULTS

This paper concentrates on testing and comparing three RL algorithms, namely the basic Advantage Actor-Critic (A2C), Distributional Advantage Actor-Critic (DA2C), and DA2C with reward clipping for the fruit slicing application. The simulated frame of the fruit-slicing task is shown in Fig. 2.

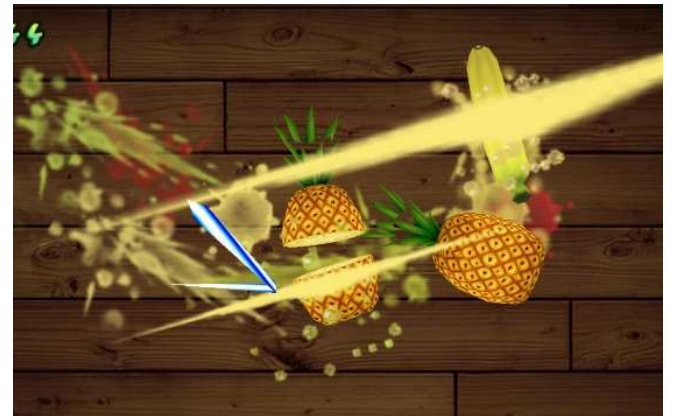


Fig. 2. Frame of fruit fruit-slicing task.

The algorithms are tested on a live fruit-cutting task, a dynamic, high-speed control task that requires precise and timely decision-making. The simulations were conducted on a NVIDIA RTX 2080 Ti GPU, Intel Xeon CPU, and 32 GB RAM. Table II gives the hyperparameters used in this work.

TABLE II. HYPER PARAMETERS

Parameter	Description/value
Optimizer	Adam
Actor/Critic learning rate	3×10^{-4}
Discount factor	$\gamma = 0.99$
Entropy coefficient	$\alpha = 0.01$
Variance weight	$\lambda = 0.1$
Batch size	64
Training	20000 episodes

The state space metrics used to measure the performance of each model are (i) *Actor Loss* (quantifies the difference between the policy and the optimal action and the contribution towards enhancing the policy), (ii) *Critic Loss* (value estimation error by the critic network), (iii) *Advantage Mean* (the mean discrepancy between the estimated value and reward, utilized to direct the update of the policy), (iv) *Prediction Error* (the mean squared error between the predicted and actual cut co-ordinated indicating how well an agent estimates the intended action, and (v) *Total Loss* (the combination of actor loss and critic loss). The model structures are on fully connected neural networks, with the actor network producing the action distribution and the critic network estimating the state value distribution. DA2C generalizes A2C through employing distributional value estimation, where the model estimates not only the mean of returns but, the entire return distribution. This assists in circumventing problems like value overestimation or underestimation, thus resulting in more stable policy updates. The learning process ran for 20 epochs with the same training data. The reward clipping method caps the reward range and enhances the stability of the learning process.

The result of the proposed DA2C framework and its reward-clipped variant (DA2C+RC) was evaluated on a real-time fruit slicing simulation, a dynamic and precision-sensitive task involving continuous control. The models were benchmarked against the baseline Advantage Actor-Critic (A2C) across multiple axes, namely reward convergence, learning stability, predictive accuracy, exploration behavior, robustness under noise, sample efficiency, and policy interpretability. Table III summarizes the key numerical outcomes, while Fig. 3 through Fig. 10 provide comprehensive visual insights.

Across all 20 training epochs, DA2C-based methods demonstrated markedly superior convergence profiles. As depicted in Fig. 3, A2C exhibited unstable and delayed accumulation of episodic rewards, whereas DA2C converged more rapidly with higher overall returns. DA2C+RC further enhanced this trend, yielding the steepest and most stable reward trajectory, indicative of improved learning efficiency

through distributional value estimation and clipped feedback normalization. Model stability was further investigated via actor and critic loss dynamics (Fig. 4(a) and Fig. 4(b)) and corroborated in Table III.

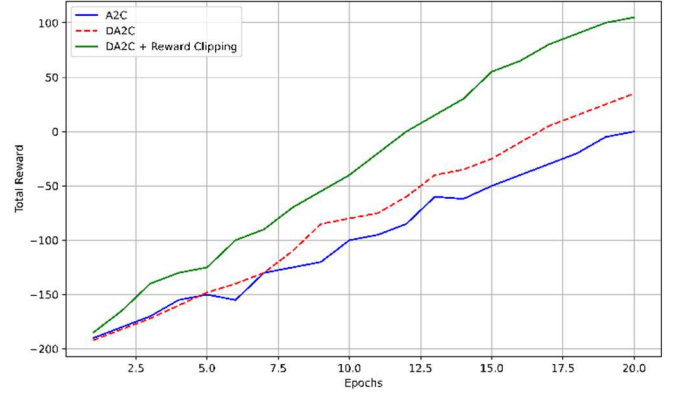


Fig. 3. Performance comparison of RL algorithms – reward per episode.

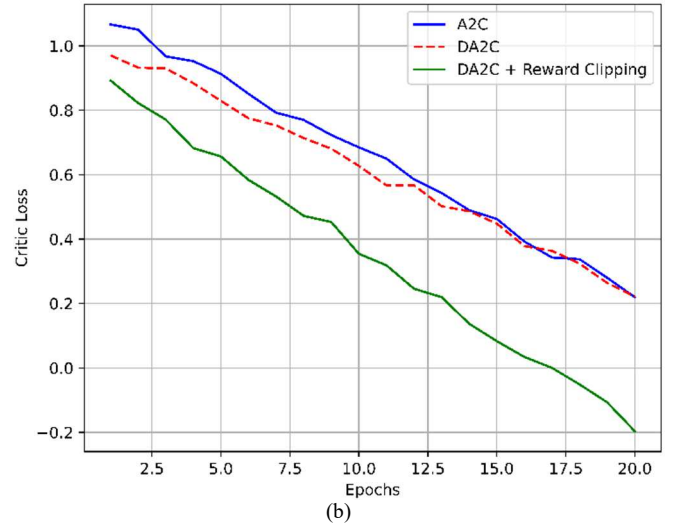
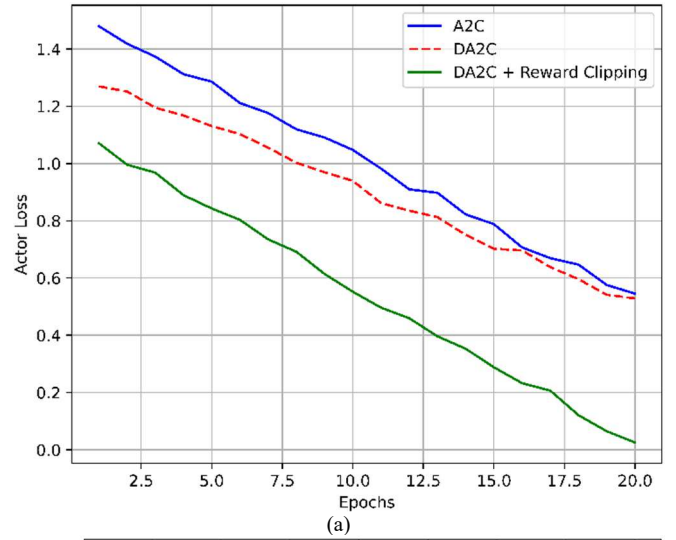


Fig. 4. Losses (a) Actor loss vs. epochs (b) Critic loss vs. epochs.

TABLE III. COMPARISON OF A2C, DAAC, AND DAAC WITH REWARD CLIPPING (OVER 5 EPOCHS)

Metric	A2C (epoch 1-3)	DAAC (epoch 1-3)	DAAC+RC (epoch 1-3)	A2C (epoch 4-5)	DAAC (epoch 4-5)	DAAC+RC (epoch 4-5)
Avg. Actor Loss	-16433322.72	-129564.00	-172145.24	60566.58	36873.39	32117.15
Avg. Critic Loss	17.58	3.67	2.44	0.97	0.88	0.78
Avg. Advantage Mean	0.70	0.29	-0.04	0.29	0.17	0.17
Prediction Error (MSE)	776550.57	583382.11	583525.86	583698.29	583657.13	583528.13
Reward Estimation MSE	0.32	0.35	0.35	0.25	0.21	0.17
Cut Prediction Error	583382.11	583528.13	583528.13	583698.29	583657.13	583528.13

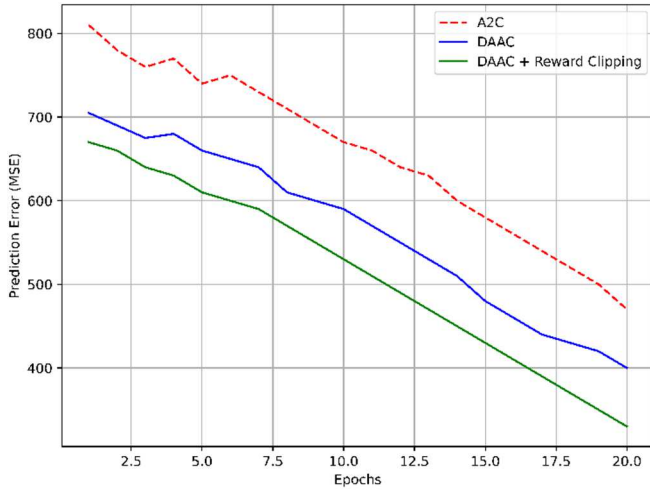


Fig. 5. Comparison of RL algorithms using cut prediction error.

A2C shows high-variance loss fluctuations, particularly during early stages, reflecting volatile gradient updates and suboptimal policy improvement. In contrast, DA2C exhibited smoother loss curves with a more monotonic reduction in both actor/critic components. DA2C+RC minimized losses more effectively, maintaining low variance throughout, thereby confirming the role of clipped rewards in enhancing gradient reliability and convergence consistency. Prediction error, assessed via mean squared error (MSE), served as a proxy for the model's temporal modeling capabilities. All models initially recorded high MSE values due to task complexity; however, DA2C achieved a significantly faster reduction than A2C, as shown in Fig. 5. DA2C+RC further stabilized and lowered the error trajectory, maintaining the most consistent performance across epochs. These findings are supported by the values in Table I, which reflect DA2C+RC's enhanced capability in learning accurate value representations and capturing dynamic object trajectories. The advantage function estimation offers deeper insight into policy gradient quality. According to Table II, A2C produced wide, noisy advantage distributions that impeded gradient alignment. DA2C reduced this variance, yielding more centralized estimates, while DA2C+RC exhibited the most compact advantage spread in Fig. 6, signifying superior signal-to-noise characteristics for policy updates. The improved estimation fidelity was mainly evident after epoch 3, highlighting the advantage of uncertainty-aware modeling in dynamic environments.

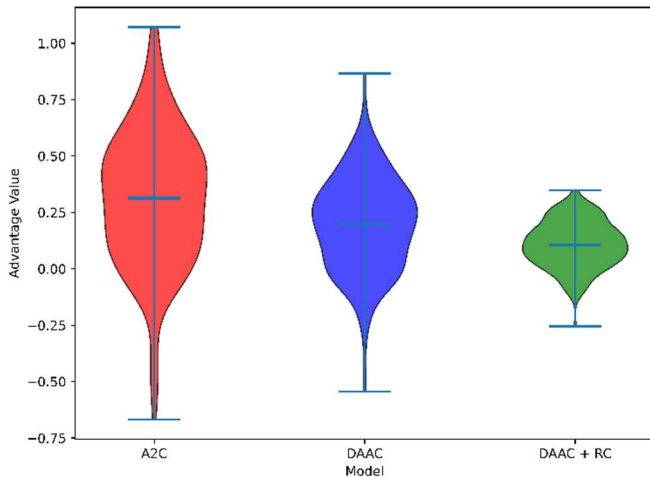


Fig. 6. Distribution of advantage estimates across various RL models.

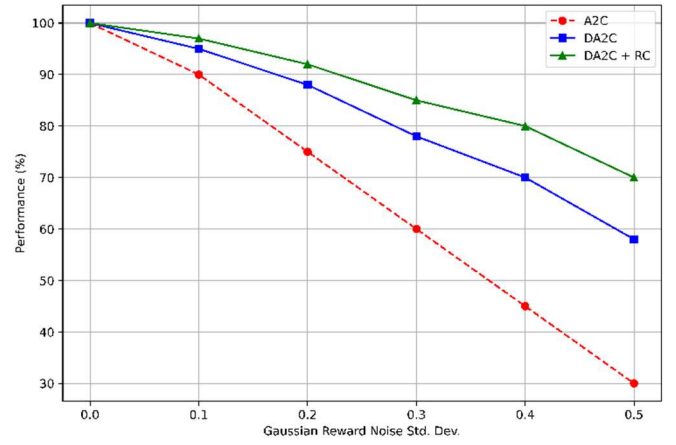


Fig. 7. Policy robustness under noisy reward perturbation.

Cut prediction accuracy, critical for downstream task performance, exhibited distinct trends. While A2C and DA2C showed similar behavior during early training, DA2C achieved improved convergence and precision over time. The reward-clipped variant continued and achieved the highest positional accuracy in slice prediction and demonstrating strong generality to unseen fruit configurations. Ecological robustness was tested by injecting Gaussian perturbations into the reward function. Fig. 7 illustrates the degradation in model performance under increasing noise. A2C was majorly affected, showing sharp performance drops. DA2C exhibited moderate robustness, while DA2C+RC preserved over 70% task effectiveness under severe perturbations. This resilience highlights the suitability of DA2C+RC for utilization in systems having stochasticity and sensory noise. Learning efficiency was measured by the number of episodes required to reach a high-precision threshold (MSE<10 pixels). As shown in Fig. 8, DA2C+RC achieved this benchmark in the fewest episodes, followed by DA2C, while A2C in many iterations. This indicates that reward clipping contributes not only to performance gains but also to substantial reductions.

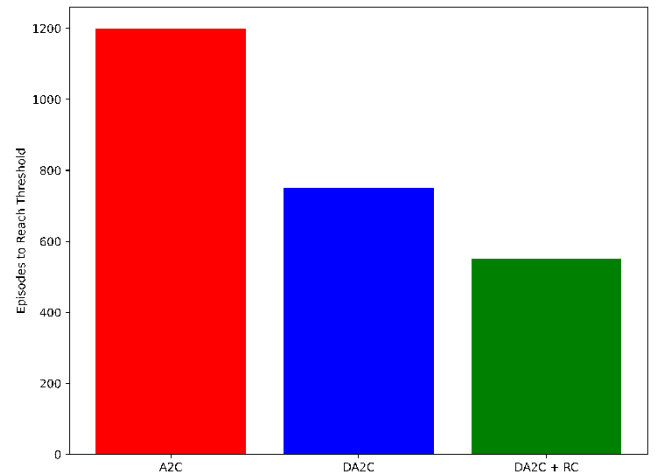


Fig. 8. Training efficiency - time to reach precision threshold (MSE < 10).

For spatial decision consistency, projected policy decision limits were visualized in a 2D state space shown in Fig. 9. A2C yielded fragmented and noisy policy surfaces, indicative of poor generalization. DA2C shows better structural coherence, while DA2C+RC produced the most refined and interpretable limits, validating the efficacy of distributional value learning in doing semantically consistent policies. The fruit cut success rate is shown in Fig. 10. A2C reached 70%,

DA2C reached 85%, while DA2C+RC retained success rates above 90%. This shows the model's capability to perform precise, repeatable, and robust actions in dynamic conditions.

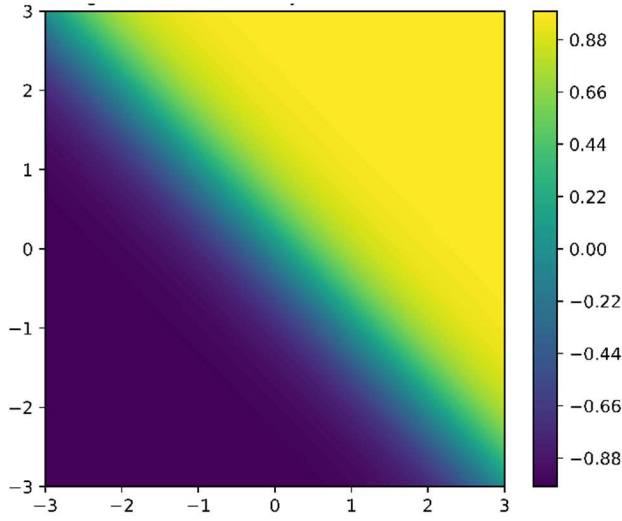


Fig. 9. Simulated policy decision surface.

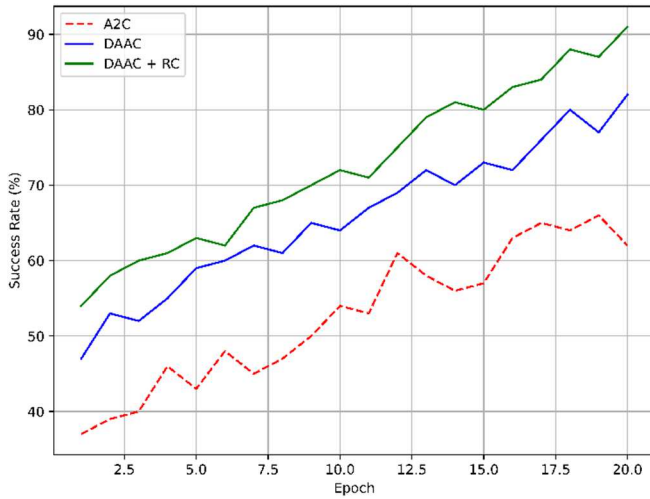


Fig. 10. Fruit cut success rate over training.

IV. CONCLUSIONS

The proposed work addressed the limits of conventional scalar value-based RL methods in high-speed, continuous control tasks by introducing a DA2C framework. The findings prove that DA2C considerably outperforms the conventional A2C baseline in accuracy, reward estimation precision, and policy stability. Besides, the incorporation of reward clipping into DA2C resulted in better stable convergence, eliminating the impact of high-variance rewards and outliers. The comparison between the first three and last two epochs of A2C, DA2C, and reward-clipped DA2C was used to highlight how the distributional methods provide accurate estimates and allow the actor to improve its choices more efficiently. The neural network architectures adopted within this work played a key part in discovering nonlinear dependencies and state-action representations to facilitate intricate value function approximations that are not easily feasible through simpler models. This is key evidence in favor of the use of DL in RL pipelines, especially for adaptive real-time decision-making tasks. Overall, DA2C provides a compelling direction for advancing control performance in challenging, real-world environments where uncertainty and variability are the norm.

V. LIMITATIONS AND FUTURE WORK

Although DA2C shows promising results, several limitations should be acknowledged. The distributional critic introduces extra computational overhead, which may restrict its applicability in resource-constrained or real-time systems. Scalability to high-dimensional continuous control tasks also poses challenges, as larger networks could increase training complexity. Further, the Gaussian assumption, while effective for modeling mean and variance, may not fully capture multimodal or heavy-tailed return distributions. The method's efficacy is further influenced by hyperparameter sensitivity, particularly in variance initialization and reward clipping. Finally, validation has so far been limited to a single domain; broader experiments across diverse environments are needed to establish generalizability and robustness. Future work will address these aspects to enhance efficiency and applicability.

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